

INST327 – Assignment 5
(UPDATED as of 4/10/2024)
Includes an alternate Question 2

NOTE: You may complete either the version of Question 2. You DO NOT need to complete both versions. Don't forget to complete Question 1 and submit your answers to 2 questions as a single file.

Please import the version of the iSchool database attached to this assignment. The script, ischool_v2.sql, attached in the assignment, will re-create the iSchool_v2 database with a clean refresh after changes from the CRUD assignment earlier this semester. Use iSchool_v2 for this assignment.

Be sure you complete both questions.

Please consult the rubric and instruction sheets before starting the assignment.

Q.1) Aggregate Functions (50 points)

Replicate the result set below which finds the maximum course duration for each course group with a unique combination of course prerequisites, delivery types, and meeting days. Investigate the result set shown and the base tables in the ischool_v2 database to identify the conditions and structure you will need in your query.

Hint: You may need to use the TIMEDIFF function to properly calculate the max_course_duration column values.

course_prereq	delivery_type	meeting_days	max_course_duration
INST631, INST632, INST717	Face-to-Face	MTh	03:45:00
INST631, INST632, INST717	Face-to-Face	MW	03:45:00
C- (INST201/301) and C- (INST126, INST327, ...	Face-to-Face	Th	02:45:00
C- (INST201/301) and C- (INST126, INST327, ...	Face-to-Face	W	02:45:00
permission	Face-to-Face	M	02:45:00
C- (INST126, MATH115, STAT100, PSYC100)	Face-to-Face	M	02:45:00
everything	Face-to-Face	W	02:45:00
everything	Face-to-Face	Th	02:45:00
INST631 & INST710	Face-to-Face	W	02:45:00
INST630/INFM603	Face-to-Face	Tu	02:45:00
INST604	Face-to-Face	W	02:45:00

11 rows returned.

Q.2 – Version 1) Aggregate Functions (50 points)

Replicate the result set below which finds the average zip code for each country and department with a unique number of people.

number_of_people	average_zipcode	country	department
2	20707	United States	MIM
9	21567	United States	ISCHOOL
1	76100	France	ISCHOOL
1	96500	Brazil	HCIM
1	21401	United States	HCIM
5	96948	Colombia	BSIS
3	234766	China	BSIS
2	175556	Ecuador	BSIS
2	76666	Egypt	BSIS
1	40198	Indonesia	BSIS
1	57000	Dominican R...	BSIS
1	43210	Thailand	BSIS
1	744000	Turkmenistan	BSIS
1	200211	Nigeria	BSIS
1	94150	France	BSIS

15 rows returned.

Q.2 – Version 2) Aggregate Functions (50 points)

Replicate the result set below which finds the largest and smallest zip code for each country and department with a unique number of people.

number_of_people	highest_zipcode	lowest_zipcode	country	department
3	22206	14201	United States	PHD
5	22206	10005	United States	MLIS
2	21401	20013	United States	MIM
1	879	879	Japan	MIM
9	39056	10451	United States	ISCHOOL
1	76100	76100	France	ISCHOOL
1	5001	5001	Albania	HCIM
1	96500	96500	Brazil	HCIM
1	8900	8900	Denmark	HCIM
1	21401	21401	United States	HCIM
5	54020	111156	Colombia	BSIS
3	54000	300000	China	BSIS
2	180110	171002	Ecuador	BSIS
2	81712	71621	Egypt	BSIS
1	40198	40198	Indonesia	BSIS
1	57000	57000	Dominican Republic	BSIS
1	43210	43210	Thailand	BSIS
1	2209	2209	Peru	BSIS
1	5006	5006	Algeria	BSIS
1	20000	20000	Brazil	BSIS
1	744000	744000	Turkmenistan	BSIS
1	3100	3100	Bangladesh	BSIS
1	200211	200211	Nigeria	BSIS
1	94150	94150	France	BSIS
1	6500	6500	Turkey	BSIS

25 rows returned.